House Price Prediction: Project Recap

Predicting house prices using the **Kaggle House Prices: Advanced Regression Techniques** dataset — exploring how linear, nonlinear, and combined modeling approaches affect prediction performance.

Original Scope

Standard comparison of regression models: Ridge, Lasso, and nonlinear methods

Refined Question

When are linear models insufficient, and does combining models improve performance?

Core Hypothesis

Ensemble methods (hybrid & stacking) improve accuracy by combining complementary model strengths

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Algorithms & Pipeline

Preprocessing

- Median imputation & categorical handling
- One-hot encoding + ordinal mapping for quality features
- Log transform: **$y = \log(1 + \text{SalePrice})$**
- Predictions back-converted: **$\hat{y} = \exp(\hat{y}) - 1$**

Models Implemented

→ Ridge Regression

$$\min \|y - Xw\|^2 + \lambda\|w\|^2$$

→ Lasso Regression

$$\min \|y - Xw\|^2 + \lambda\|w\|_1$$

→ Gradient Boosting

$$F_m(x) = F_{m-1}(x) + \gamma_m h_m(x)$$

→ Hybrid & Stacking

$\hat{y}_{hybrid} = \alpha \cdot \hat{y}_{ridge} + (1-\alpha) \cdot \hat{y}_{GB}$; Stacking via meta-learner $f(\hat{y}_{ridge}, \hat{y}_{GB})$

All models evaluated with **5-fold cross-validation** and **RMSE**.

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Preliminary Results

~29k

GB RMSE

Best performer — Gradient Boosting

~51k

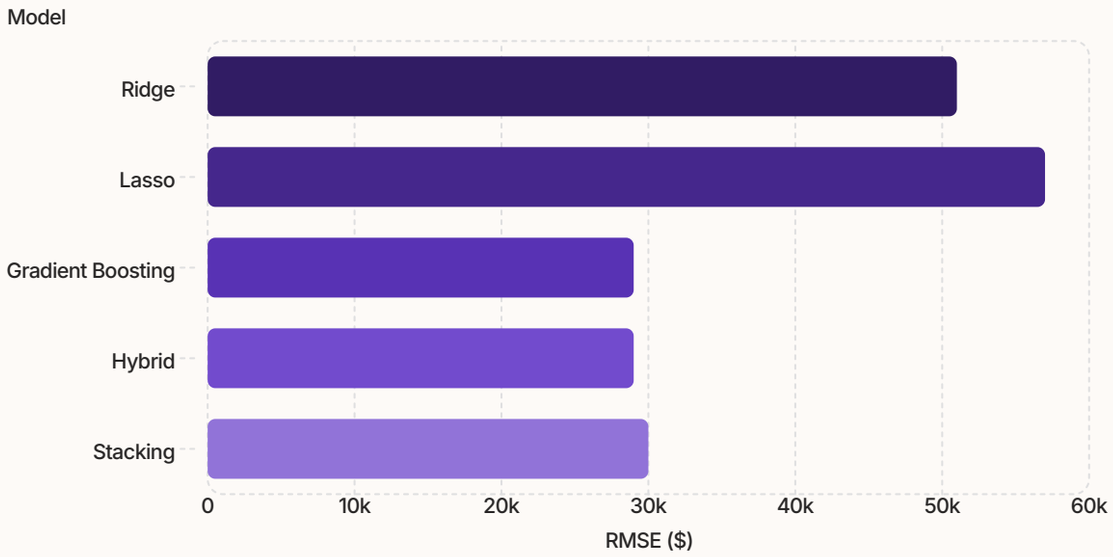
Ridge RMSE

Linear baseline

~57k

Lasso RMSE

Highest error among all models



Hybrid ≈ GB and Stack ≈ GB — ensemble methods did not improve over Gradient Boosting alone, indicating the base models are not sufficiently complementary.

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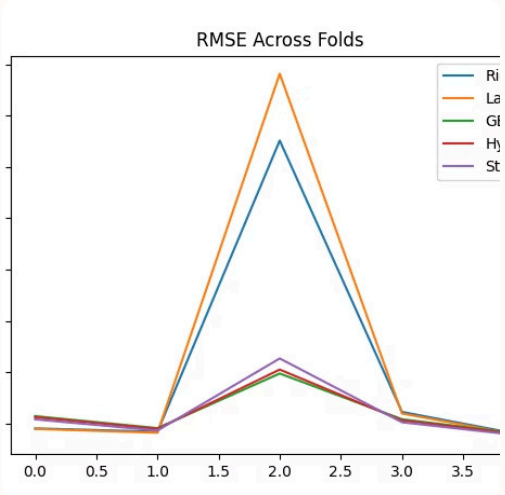
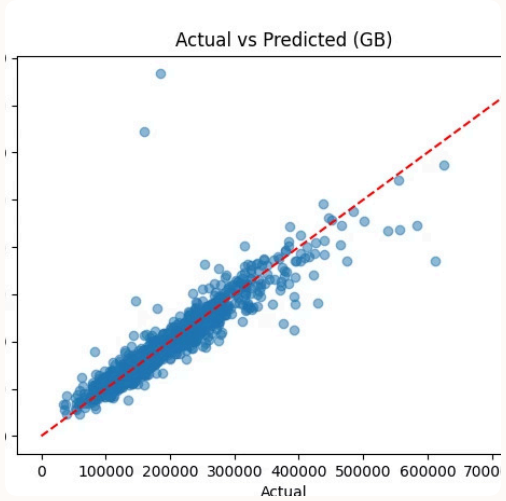
Graph Interpretations



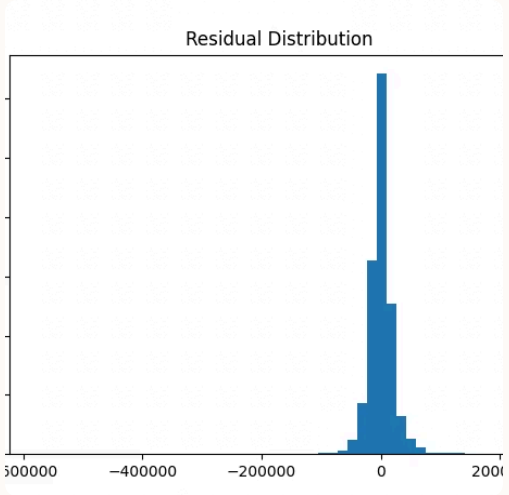
RMSE Across Folds

GB is consistently lower across all folds. Ridge & Lasso show large spikes — indicating instability and underfitting.



Actual vs. Predicted (GB)

Points align closely along the diagonal — accurate overall. Slight spread at higher prices reveals larger errors for expensive houses.



Residual Distribution

Centered around 0 — model is unbiased. Most residuals within ±30k, confirming RMSE ≈ 29k. Some large outliers increase RMSE.

Error vs. Price

Clear pattern: higher price → larger error. Suggests **heteroscedasticity** — model struggles most with expensive houses.

Top features by importance: **OverallQual, GrLivArea, TotalBsmtSF** — matching real-world intuition and confirming the model learns meaningful patterns.

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Challenges & Solutions

Scope Too Simple

Initial design was a basic model comparison → expanded to hybrid + stacking to study model complementarity.

Complex Preprocessing

Handled via a structured pipeline: imputation, encoding, log transformation, and ordinal mapping.

Linear Models Underperformed

Ridge and Lasso showed high, unstable RMSE → introduced Gradient Boosting as the primary nonlinear model.

Hybrid Model Failed Initially

Residual-based hybrid did not work → switched to weighted hybrid: $\hat{y} = \alpha \cdot \hat{y}_{ridge} + (1-\alpha) \cdot \hat{y}_{GB}$.

Data Leakage Risk

Avoided by not tuning α dynamically on validation data.

Stacking Did Not Improve

Revealed lack of model complementarity — GB already captures both linear and nonlinear relationships.

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Final Report Plan & Next Steps

Final Report Will Include

- Finalized comparisons across all models
- Improved explanation of hybrid and stacking approaches
- Strengthened graph interpretation
- Clear connection of results back to hypothesis

If Time Permits

01

Explore LightGBM / XGBoost

More powerful boosting alternatives to test further gains

02

More Diverse Stacking

Improve ensemble with additional base models for true complementarity

03

Feature Engineering

Create new features to improve signal quality

04

Error Analysis by Price Range

Address heteroscedasticity and improve accuracy for expensive houses

Key Takeaway: Gradient Boosting already captures both linear and nonlinear relationships, leaving little room for ensemble gains — a meaningful finding in itself.

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